Project 1 Report

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1. Approach

My approach to this project was when looking at the key points in images, for each distinctive feature of the image, like (car wheels, lines, etc.). I noticed that there was a cluster of points around it, so my approach was to cluster those distinctive groups into one cluster and analyze what type of geometrical structure it has and run an appropriate algorithm to create edges.

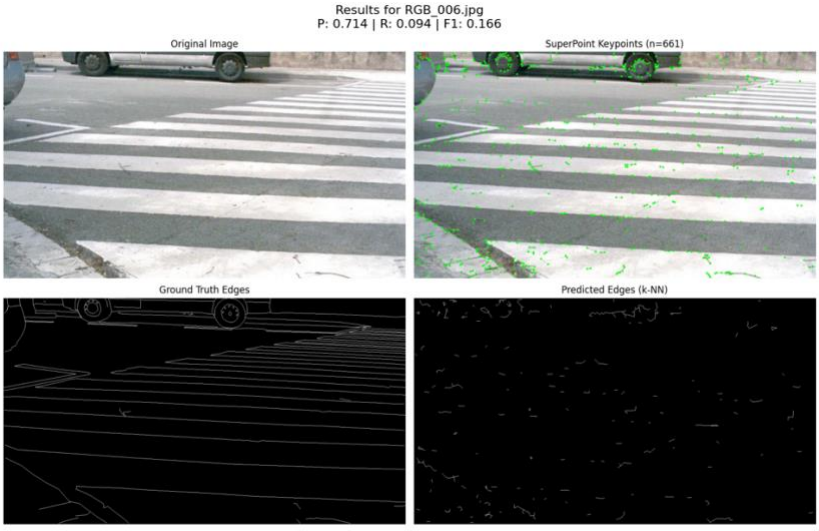
2. Design Decision

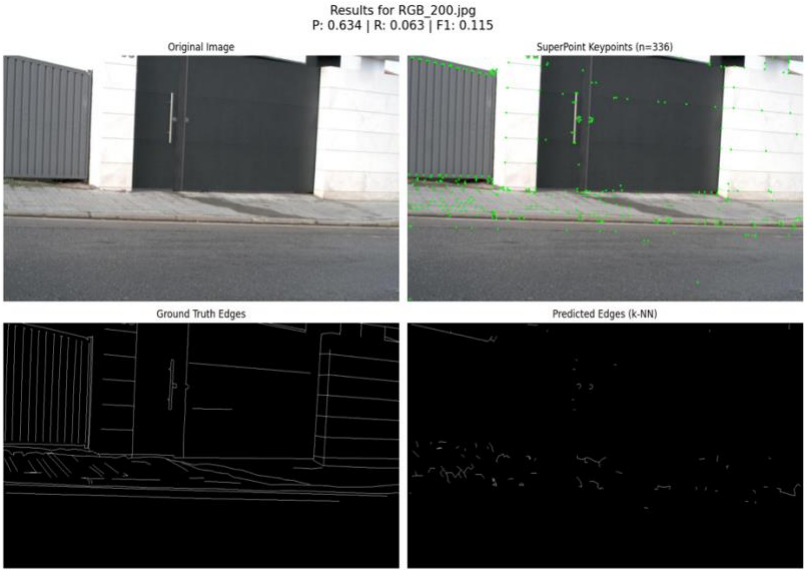
First, I used DBSCAN instead of k-mean clustering since DBSCAN can group them in a more complex shapes instead of spherical shape like k-mean. Then on those clusters we use PCA to figure out if cluster is linear structure like (roads, lines, etc.) or circular/compact like (wheels, signs etc.). And if the algorithm determines if its linear, we use Kruskal's algorithm to determine minimum spanning tree and if its circular/compact we use quickhull algorithm to find convex hull. And since there must be some noise in the shape, we filter them out by SuperPoint descriptor where we compute cosine similarity between 2 descriptors then we skip if the shape is Convex Hull like wheels, and if the cosine similarity is < 0.5 then line is erased. And for final iteration we connect endpoints if they are, 1. close, 2. pointing at each other and 3. having matching descriptors.

3. Failure Cases

In regions with low texture (smooth walls, shadows, etc.), SuperPoint detects very few key points. Our algorithm (MST/Connect Gaps) cannot reconstruct an edge if the underlying nodes are missing. Ex) RGB_006

On highly textured surfaces (gravel, foliage, etc.), the "descriptor similarity" filter can be too aggressive, rejecting valid edges because the pattern changes rapidly. Ex) RGB_200





4. Quantitative Results

Average

Precision	0.764	The does good job of avoiding false positives
Recall	0.281	Misses edges where key points are spread out
F1 Score	0.398	
Background Accuracy	0.997	

5. Comparison to Naïve Baseline

Metric	Proposed Model	Naïve Baseline using drawing disk around a point
Precision	0.764	0.747
Recall	0.281	0.665
F1	0.398	0.693
Background Accuracy	0.997	0.987